✏️ RS004 Week 10 — English Worksheet

Topic: Paper to Code (Apply) · **Course:** Platformer Game · **Time:** about 45 minutes

This week you design a level **on paper first**, then turn it into coordinates and into code. You will measure how far and how high your player can jump, so every platform is actually reachable.

> Keep these words handy: **sketch**, **coordinate (x, y)**, **jump height**, **jump distance**, **reachable**.

1 · Predict 🔮

Read each snippet. Before you run it, write what you think will happen.

```
# JUMP_STRENGTH = -14, GRAVITY = 0.7
# max jump height ≈ 14² / (2 × 0.7) = 140 pixels
```

If your player is on a platform, roughly how high can the next platform be and still be reachable?

```
MY_LEVEL = [
    (0, 580, 250, 20),
    (350, 520, 100, 20),
]
```

The first platform ends at x = 250. The next starts at x = 350. How wide is the gap to jump?

```
START_X, START_Y = 50, 540
FINISH_X, FINISH_Y = 450, 220
```

Where does the player begin, and where is the goal? Lower `y` is higher on screen.

2 · Measure Your Jump 📏

Before you design, find out your player's real limits. Run your week 9 game and test.

```
1. Stand on the ground and jump straight up.
   Highest platform you can land on is about ___ pixels above.


2. Run and jump across a flat gap.
   Widest gap you can clear is about ___ pixels.
```

Why must you know these numbers *before* you place platforms?

3 · Spot the Bug 🐛

Each snippet is a *level design* mistake. Read what's wrong, then rewrite the coordinates so the level works. Explain your fix.

Bug A — The second platform is meant to be reachable, but it is 300 pixels above the first — too high to jump.

```
MY_LEVEL = [
    (0, 580, 250, 20),
    (300, 280, 100, 20),
]
```

Hint: max jump height is about 140 pixels. Lower the gap.

Write the fixed coordinates:

Why was it wrong? Why does your fix work?

Bug B — The finish marker is floating in empty space with no platform under or near it, so the player can never reach it.

```
MY_LEVEL = [
    (0, 580, 250, 20),
    (350, 520, 100, 20),
]
FINISH = (700, 100, 50, 40)
```

Hint: add a platform near the finish, within jump range of the last one.

Write the fixed level:

Why was it wrong? Why does your fix work?

4 · From Sketch to Coordinates 🛠️

Do this in order.

1. On paper, sketch a level: a start, 5–8 platforms, and a finish at the top. Take a photo when done.

2. For each platform in your sketch, write its `(x, y, width, height)`. Read coordinates off your drawing.

3. **Stretch:** add a secret platform reachable only by a hard jump. Where is it?

5 · Build & Show 📸

Turn your paper sketch into a playable level. Tune it so it is challenging but fair — aim for about 1 success in 5 tries.

Send **your paper sketch photo and a full play-through video**, then explain what you did. Use these starters — write 4 to 6 sentences.

> First, I sketched my level with …
>
> I measured my jump and found I can reach about ___ pixels high and ___ wide.
>
> To turn the sketch into code, I …
>
> The hardest jump in my level is …
>
> One tricky moment was …
>
> If I had more time, I would …

6 · Record Your Walkthrough 🎥

Take a video of a full clear of your hand-designed level. Teach it to someone new. Try to use these words: **sketch**, **coordinate**, **jump height**, **reachable**, **difficulty**.

1. Show your paper sketch next to the running game.
2. Point at one platform and read its coordinates.
3. Show the hardest jump and whether you cleared it.
4. Reach the finish.

Write what you will say in your video. Plan it here first.

Submit ✅

Send this worksheet + your paper sketch + your walkthrough video to teacher on KakaoTalk.